



**ALPHA CHOICE INNOVATIVE ACADEMY,
AREPO**

GRADE 8

THIRD TERM

PHYSICS

MASTERNOTE

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SYNOPSIS	
WEEKS	TOPICS
1-2	SOUND
3-4	MAGNETISM
5	NON METALS (HYDROGEN AND OXYGEN)
6	MID-TERM TEST
7	MID-TERM BREAK
8	WATER
9-10	RADIOACTIVITY

WEEK 1\$2: SOUND

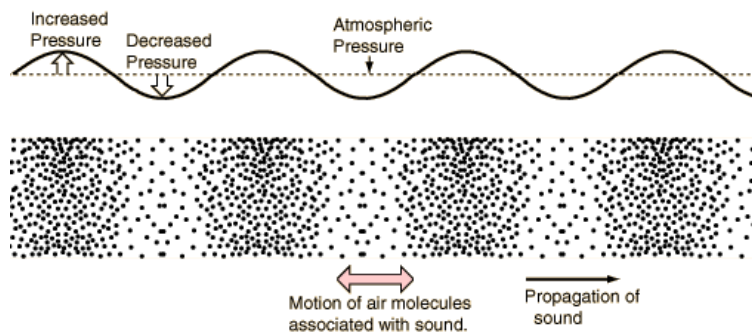
Objectives: By the end of the lesson, the students should be able to:

- i) explain how sound waves travel
- ii) describe the process of sound detection by the ear
- iii) relate loudness and amplitude
- iv) relate the pitch of a sound to the frequency
- v) explain the Doppler shift and state its uses.

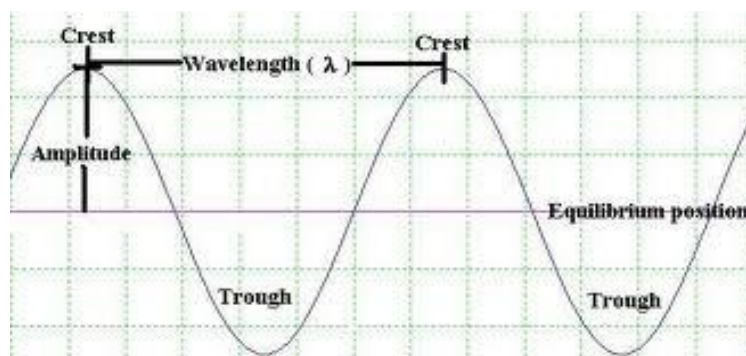
Any object makes a sound when it vibrates. A vibration is a movement about a fixed point. This movement may be described as a to-and-fro movement or a backwards and forwards movement.

Sound waves can travel in a gas, a liquid or a solid because they all contain particles. When an object vibrates, it makes the particles next to it vibrate too. For example, when an object vibrates in the air, it pushes on the particles around it.

As the vibrating object moves towards the air particles, it squashes them together. The particles are not compressed but the pressure in the air at that place rises because the particles are closer together. As the object moves away from the particles next to it, it gives them more space and they spread out and the pressure at that place falls.



If these changes in pressure are plotted on a graph, they make a waveform similar to that shown below.

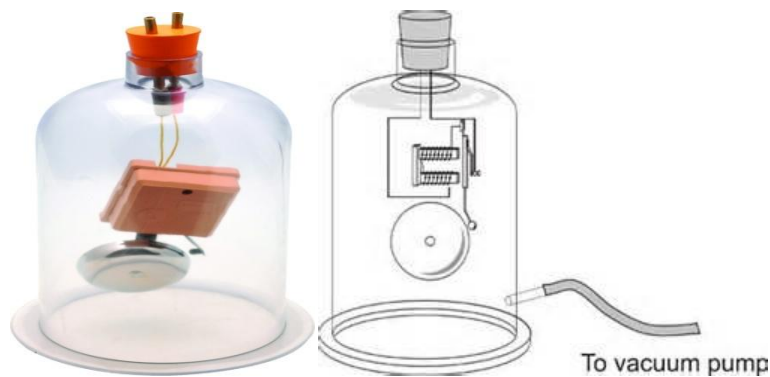


Sound waves are generated and travel in liquids and solids in the same way they do in gases. The particles in liquids and solids are held together by forces of attraction. In a liquid, the particles are further apart than they are in a solid and can move around one another. Sound therefore travels very well through a liquid. It moves faster and further than it does in a gas.

When sound travels through a solid, it moves even faster than when it travels through a liquid because of the close interaction of the particles. However, the sound does not travel so far. **In general, the velocity of sound is greater in solids and liquids than in gases.**

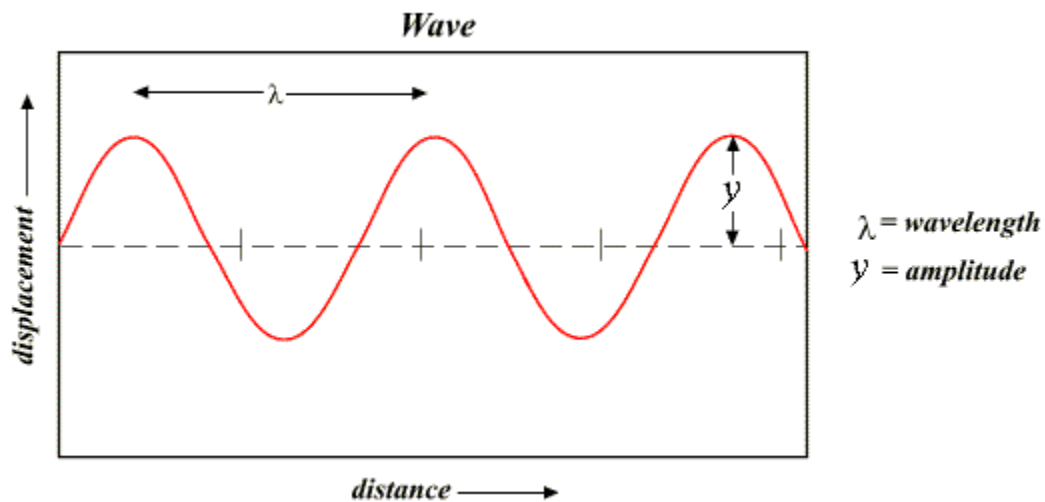
A humpback whale emits a series of sounds called songs which it uses to communicate with other whales. The sound travels thousands of kilometers through the ocean. A snake detects vibrations in the ground with its lower jaw bone. The bone transmits the vibrations to the snake's ears and allows the snake to detect the footsteps of its prey.

Sound waves cannot pass through a vacuum because it does not contain any particles. As air is drawn out of the bell jar with a pump, the sound of the bell becomes quieter. When a vacuum is established in the bell jar, the bell cannot be heard though the hammer can be seen striking the bell.



Describing a wave

The **amplitude** of a wave is the height of the crest or the depth of the trough and shows the maximum displacement of the particles from their rest position. The **wavelength** is the distance from the top of one crest to the top of the next crest or from the bottom of one trough to the bottom of the next trough.

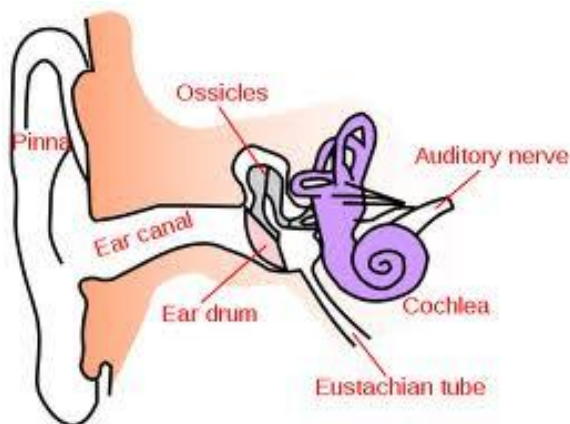


Sound detection

Sound detectors convert sound energy into another form of energy. In a sound detector, a diaphragm vibrates at the frequency of the sound wave. The vibration of the diaphragm is then converted into another form of energy. A microphone is a device that converts sound energy into electrical energy.

The ear is an amazing sound detector. It not only detects sound waves over a very wide range of frequencies but it is also sensitive to an enormous range of wave amplitudes. Human hearing can distinguish many different qualities of sound.

The ear is not equally sensitive to sound of all frequencies. Most people cannot hear sounds with frequencies below 20 Hz or above 20 000 Hz. In general, most people are most sensitive to sound with frequencies between 1000 Hz and 5000 Hz. Older people are less sensitive to frequencies above 10 000 Hz than are young people. By 70 years of age, most people can hear nothing above 8 000 Hz. This loss affects the ability to understand speech.

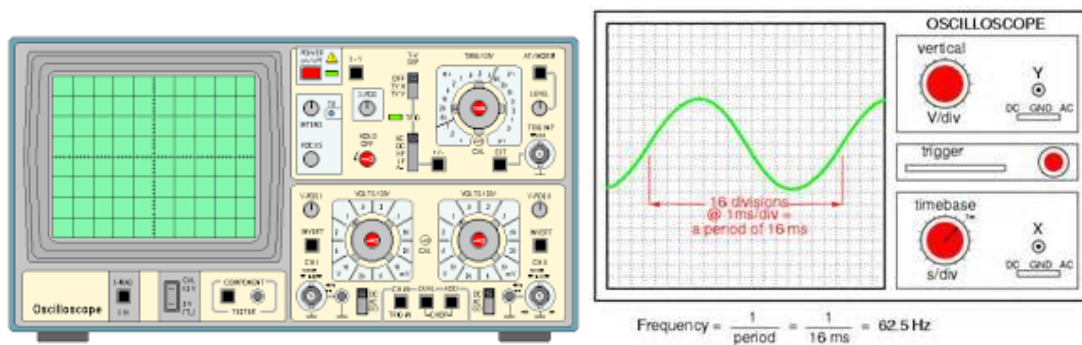


The ear is divided into three parts - the outer, the middle and the inner ear. The outer ear consists of the fleshy, visible part of the ear called the pinna, which collects sound; the auditory

canal; and the eardrum. Sound waves cause vibrations in the eardrum. The middle ear consists of three tiny bones in an ear- filled space in the skull. The bones transmit the vibrations of the eardrum to the oval window on the inner ear. The inner ear is filled with a watery liquid. Sound vibrations are transmitted through the liquid into sensitive portions of the spiral-shaped cochlea. In the cochlea, tiny hair cells are vibrated by the waves. Vibrations of these cells stimulate nerve cells that lead to the brain, producing the sensation of sound. *If the sound is very loud when it enters the ear, the vibrations are so strong they damage the fibres and the nerve endings are not stimulated. This can give rise to permanent damage and deafness.*

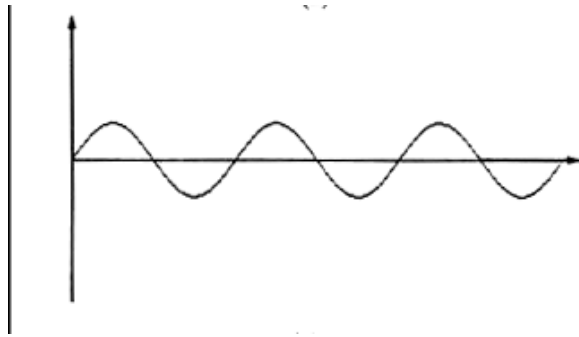
Over exposure to loud sounds can cause the ear to lose its sensitivity, especially to high frequencies. The longer the exposure, the greater the effect. Short term exposures could clear in a period of hours but long-term exposure could take days or weeks to be recovered from. Long exposure to 100dB or greater sound levels can produce permanent damage. Specifically designed ear muffs and inserts can reduce the sound and its effects to the ear much better than cotton earplugs and special ear inserts. Another source of hearing loss is the result of loud music from stereo headphones on personal radios and tape players. Some doctors have said that an earphone is “like the nozzle of a fire hose stuck down the ear canal”.

The instrument which allows one to see the waves produced by the sounds is called an oscilloscope. There are two types: the cathode ray oscilloscope (CRO) which is used in some school laboratories and the digital oscilloscope which does the same work as the CRO but only differs in that displays produced on the screen can be stored.

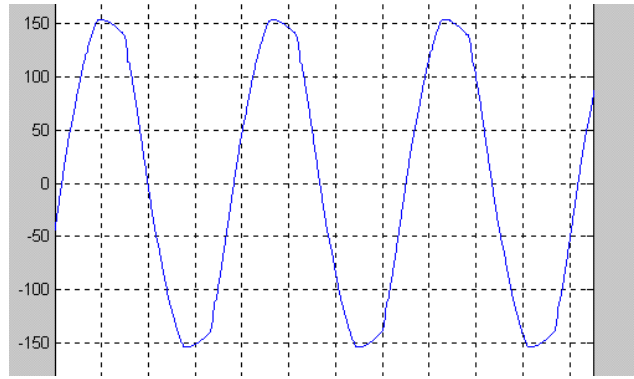


The loudness of a sound

The loudness of a sound is related to the movement of the vibrating object. If an object only moves a short distance from its rest position, it will produce a sound wave with only a small amplitude and the sound that is heard will be a quiet one. If the object’s distance from its rest position is large, it will produce sound waves with a large amplitude and the sound heard is a loud one. The loudness of a sound is measured in decibels.



Quiet sound



loud sound

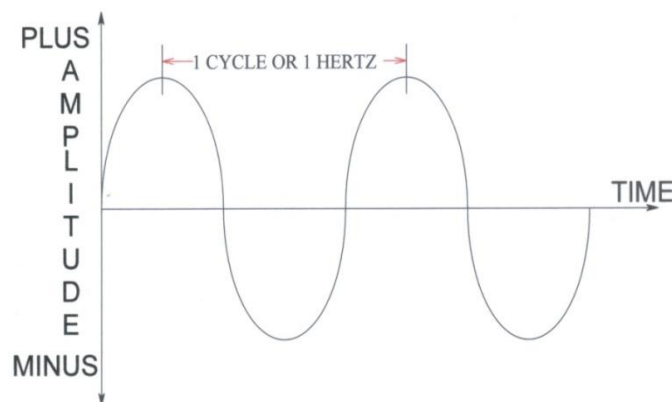
Effect of loudness on the ear The vibrating air particles of a very loud sound can produce such a strong pushing and pulling force that the eardrum could have a hole torn in it. Thus, the eardrum is said to be perforated. Its vibration is no longer efficient and the person loses his or her hearing. The eardrum can heal and normal hearing can be restored.

When a person is exposed to a very loud sound or a particular note for a long time, there will be a permanent damage to a nerve ending in the cochlea and the person will no longer be able to hear.

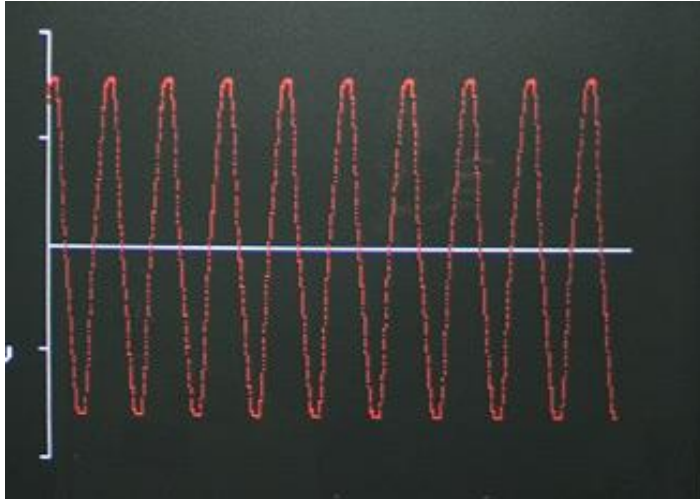
People who perform in pop groups are at risk of developing this kind of deafness, called nerve deafness. With time, they can't hear a range of notes which they used to play frequently. Also, people who work in noisy surroundings like in airports and metal factories wear ear muffs to protect their ears. It covers their ear and reduces the amount of sound energy entering the ears.

Some people have temporary deafness due to the development of ear wax in the outer ear. It can be removed under the supervision of a nurse. Also, some people have growths of tissue in their middle ear which does not allow the ear bones to move freely. People with this may be prescribed hearing aids.

Pitch of a sound The pitch of the sound an object makes depends on the number of sound waves it produces in a second. The number of waves per second is called the frequency. The frequency of a sound is measured in hertz (Hz). The higher the frequency of the wave, the higher the pitch of the sound.



Low-frequency sound (low pitch)

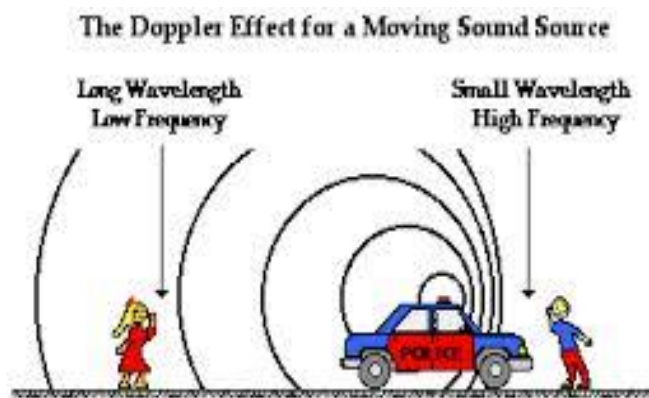


High-frequency sound (high pitch)

It is worthy of note that higher-frequency waves have a shorter wavelength than the lower-frequency waves. Sound waves share this property with light waves. The ear of a young person is sensitive to frequencies in the range of 20Hz - 20 000 Hz (Hertz) but the ability to detect the higher frequencies decreases with age. Some may have a restricted range of hearing due to nerve damage. They may be able to hear some low-pitched or high-pitched sounds.

THE DOPPLER SHIFT

Have you ever noticed the pitch of an ambulance, fire or police siren as the vehicle sped past you? The frequency is higher when the vehicle is moving towards you, then suddenly drops to a lower pitch as the source moves away. This effect is called the Doppler shift.



When the sound source is moving towards you, it emits waves which spread in circles around the location of the source at the time it produced the wave. The frequency of the sound source does not change, but when the source is moving towards the sound detector, more waves are crowded into the space between them. The wavelength shortens. Because the velocity does not change, the frequency of the detected sound is greater. When the source is moving away

from the detector, the wavelength is lengthened and the detected frequency is lower. A Doppler shift also occurs if the detector is moving and the source is stationary. It occurs for all types of waves (mechanical and electromagnetic). It is used by:

- i) radar detectors to measure the speed of baseballs and automobiles
- ii) astronomers to measure the speed and infer the distance of distant galaxies
- iii) physicians to detect the velocity of the moving heart wall in a fetus using ultrasound
- iv) a bat to detect and catch flying insects(when an insect is flying faster than the bat, the reflected frequency is lower but when it is catching up with the insect, the reflected frequency is higher.

RESEARCH:

Briefly describe how the bat and the dolphin receive sound and how these sound receptors influence their hunting and their life styles generally.

EXERCISES

1. How do sound waves travel in solids, liquids and gases?
2. What can be used in the human body to detect sound waves?
3. Why do people put a hand to their ear when they are listening to someone who is whispering?
4. What may happen to the ears of those who wear headphones and turn up the sound?
5. What is the relationship between loudness of a sound wave and its amplitude?
6. How does the pitch of a sound relate to its frequency?
7. What kind of ear damage might be caused by a loud explosion?
8. The following are three frequencies of sound waves: 1800 Hz, 50Hz, 10 000Hz.
Which has the highest and which has the lowest pitch?
What does Hz stand for?
9. When a ringing bell is placed inside a jar connected to a vacuum pump and the air is removed, no sound is heard. Explain why.
10. Does the Doppler shift occur for only some types of waves or all types of waves?
11. A bat emits short pulses of high frequency sound and detects the echoes.
 - a) In what way would the echo from large and small insects compare if they were the same distance from the bat?
 - b) In what way would the echo from an insect flying toward the bat differ from that of an insect flying away from the bat?

WEEK 3\$4: MAGNETISM

Objective: At the end of the lesson student should be able to:

- i. Explain the behavior of magnets
- ii. Explain the composition of a magnet
- iii. Understand the concept of magnetic field
- iv. Show the link between magnetism and electricity
- v. Define electromagnet and explain its concept

Magnets are materials that attract metals, when they are suspended freely from a thread, a bar magnet align itself in a north – south direction.

Materials that are not attracted by magnets are called **non – magnetic materials** example are wood, paper, plastic human body, stone, wall etc.

At the end of a bar magnet is a place where the magnetic force is stronger than at the other places in the magnet. These places where the magnetic force is strongest are called the **poles** of the magnet. The end of the magnet which points towards north when the magnet is free to move is called the **north - seeking** pole or **North Pole**. At the other end of the magnet is the **south - seeking pole** or **south pole**

When two different (unlike) ends of a magnet, i.e (North and South poles), are brought together they attract each other. But when two similar (like) ends of a magnet i.

e (South – South or North – North) are brought together they repel each other.

Composition of a magnet

Group of particles from which a magnetic material is made form tiny regions called **domains**. Each **domain** behaves like tiny magnet. If the domains are arranged at random the material does not attract other magnetic material, but when they are arranged in line where all the north poles face in one direction likewise the south Pole, when the material is in this condition, it has been magnetized and so is now a magnet.

The magnetic field

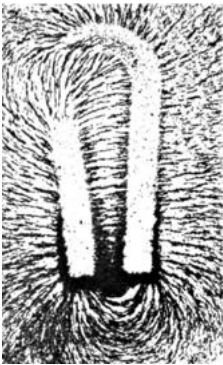
The region around a magnet in which the pull of the magnetic force acts on a magnetic material is called **magnetic field**. At the center of the earth is the earth core. It is made from iron and nickel and it is divided into two parts the inner core made up of solid metal and the outer core made of liquid metal, as the earth move the two core spins moving at different speed and generate magnetic field in the process. This makes the earth seem as if it has a large bar magnet inside it.

Relationship between magnetism and electricity

Hans Christian Oersted (1777- 1851) was a Danish scientist who studied electricity. In one of his experiments he was passing an electric current along a wire from a battery when he noticed the movement of a compass needle which had been left near the wire. This chance observation led to many discoveries about how magnetism and electricity are linked together and has many modern applications.



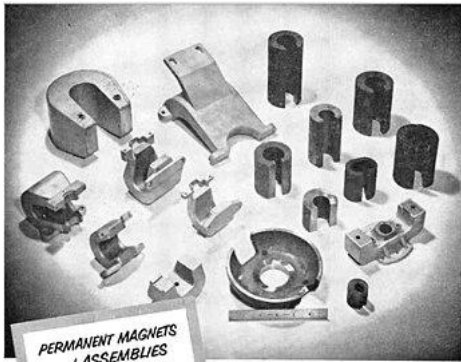
A horseshoe magnet made of [alnico](#), an iron alloy. The strong magnetic field between the poles allows the magnet to pick up a heavy piece of [iron](#).



Magnetic field of a horseshoe magnet. The field is greatest where the lines are densest, around the poles (lower)



Alnico horseshoe magnet used in a [magnetron](#) tube in an early microwave oven. About 3 in (8 cm) long.



Assortment of Alnico horseshoe magnet shapes available from a manufacturer in 1956.

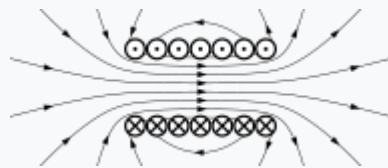
Notes

1.

1. This flux depends on the amount of magnetised material.

Part of a series of articles about

[Electromagnetism](#)



- [Electricity](#)
- [Magnetism](#)

[Electrostatics](#)[\[show\]](#)

[Magnetostatics](#)[\[show\]](#)

[Electrodynamics](#)[\[show\]](#)

[Electrical network](#)[\[show\]](#)

[Covariant formulation](#)[\[show\]](#)

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| **Magnetic Fields&Magnetic Force.** Iron filings attracted to a horseshoe **magnet** show the **magnetic field**. **Magnetism** is one aspect of the combined **electromagnetic** force. It refers to physical phenomena arising from the force caused by **magnets**, objects that produce **fields** that attract or repel other objects.Jul 29, 2015

When electric current passes through a wire it generates magnetic field around the wire. A compass can be placed at different positions on a card around the wire and the line of force can be plotted. **Lines of forces do not only show the direction as given by the plotting compass but also the strength of the field in different places. The lines of force are close together where the field is strong and further apart where the field is weaker.**

If the wire is made into a coil and connected into a circuit, a magnetic field is produced around the coil. **If a piece of steel is placed inside the wire coil and the current is switched on, the magnetism of the coil and the steel is stronger than that of the coil alone. The current flowing in the coil induces magnetism in the steel. When the current is switched off, the steel keeps some of the magnetism it acquired because it is magnetically hard**

If a piece of iron is placed inside the coil it makes an even stronger magnet when the current is switched on than the steel did. When current is switched off, the iron loses its magnetism completely because it is magnetically soft.

Electromagnet

An electromagnet is made from a coil of wire surrounding a piece of iron. When a current flows through the coil, magnetism is induced in the iron, the coil and iron form a strong electromagnet. When the current is switched off, the electromagnet loses its magnetism completely.

Uses of electromagnets

- It is used in a scrap yard to move the steel bodies of cars
- Electric bell

The electric bell

The diagram bellow shows the workings of an electric bell, when the switch is pushed the current passes through the coil and the electromagnet pull the armature to it. This makes the hammer strike the gong. When the armature is pulled to the electromagnet a gap develops between the springy metal strip and the contact screw and the circuit is broken. The current stops flowing and the electromagnet loses its magnetism. This makes the armature swing back to its original position. The springy metal strip and the contact screw now touch again and complete the circuit so the armature is pulled to the electromagnet once more. The bell is made to ring by the repeated beating of the hammer until the push switch is released.

Assignment

1. If you cut a magnet in half does each half become a magnet? Explain your answer
2. A piece of steel can be made into a magnet by repeatedly stroking it with a magnet. How does this affect the domain?
3. A nail is magnetized by being in contact with one end of a magnet. Can it still attract magnetic materials to it when it is no longer in contact with the magnet? Explain your answer.

MULTIPLE CHOICE QUESTIONS:

1. Any material that doesn't attract metal to itself is called_____
 - a. Magnetic material
 - b. Semi-magnetic material
 - c. Non-magnetic material
 - d. Insulator
2. A magnet has _____ number of poles
 - a. 1
 - b. 2
 - c. 3
 - d. 4
3. Like poles of a bar magnet will attract each other.
 - a. True
 - b. False
 - c. Maybe
 - d. I don't know
4. Which of the following statement is true for a magnet
 - a. Group of particles from which a magnetic material is made form tiny regions

- b. These tiny regions are made of domains
 - c. Each domains behaves like a tiny magne
 - d. All of the above.
5. Any object that does not produce light by itself is called _____
- a. Opaque object
 - b. Luminous object
 - c. Non- luminous object
 - d. Transluscent object
6. A region where a force is experienced is called _____
- a. Gravity
 - b. Force
 - c. Field
 - d. Magnetism
7. The distance through which light travels is called _____
- a. Ray
 - b. Beam
 - c. Reflection
 - d. Refraction
8. A ray that strikes a surface is called _____
- a. Reflected ray
 - b. Deflected ray
 - c. Incident ray
 - d. None of the above
9. There are _____ types of images formed with light
- a. 1
 - b. 2
 - c. 3
 - d. 4
10. Group of particles from which a magnetic material is made form tiny regions called _____
- a. Magnet
 - b. Dipole
 - c. Domain
 - d. Electromagnet

WEEK 5: NON-METALS

Hydrogen

Objective: At the end of the lesson, student should be able to:

- Explain how hydrogen can be prepared in the laboratory
- State the properties of hydrogen
- Explain how to identify hydrogen gas
- Give uses of hydrogen

Air is a mixture of gases such as oxygen, carbon (iv) oxide, nitrogen, noble gases and water vapour. All these gases are important to man. However, there is a gas that is not part of these components of air because it is lighter than air. When this gas evolves on the earth, it rises and settle above the air, yet the gas is useful in many ways to man. This gas is **hydrogen**.

Laboratory preparation of hydrogen gas

There are two ways to prepare hydrogen in the laboratory

- Preparation of hydrogen using a metal and an acid
- Preparation of hydrogen by action of steam on metal

Properties of hydrogen

- It is a colourless gas, that is, it has no colour
- It is odourless and tasteless
- Hydrogen is lighter than air. It is the lightest known gas to man
- It can be collected in liquid form
- It does not support burning, but it burns lighted taper as evidence that it allows things to burn in it. When hydrogen burns in air, it does so with a hot, blue flame.
- When hydrogen is burnt in oxygen, it explodes and gives a pop sound

Uses of hydrogen gas

- **As fuel:** Liquid hydrogen is used in some space rockets
- **Filling of balloons:** Hydrogen can be used to fill meteorological balloon because it is very light
- **Preparation of ammonia:** Hydrogen reacts with nitrogen gas to produce ammonia gas. This gas can be compressed to a liquid used as refrigerant in mortuaries
- **Production of margarine:** in industries, hydrogen is used with vegetable oil to make margarine.

Oxygen

Objective: At the end of the lesson, student should be able to:

- Explain how oxygen can be prepared in the laboratory
- State the properties of oxygen

- Explain how to identify oxygen gas
- Give uses of oxygen

As leant earlier, air is a mixture of gases. One of them is $\frac{1}{15}^{\text{th}}$ of air and is very useful in many ways such as burning and breathing. This gas can be found both in an uncombined state in the atmosphere, and in a combined state in oxides of metals and non metals. The gas is called **Oxygen**.

Laboratory preparation of Oxygen

Oxygen can be prepared in two major ways in the laboratory

- Preparation from hydrogen peroxide
- Preparation from potassium trioxochlorate (iv)

Properties of oxygen

- Oxygen is colourless, odourless and tasteless
- It is a gas at room temperature
- Oxygen support burning. The burning of oxygen in air is a process called combustion
- Oxygen makes up 21% of air. It's a neutral gas i.e not acidic or basic
- It rekindles a glowing splint

Oxides

Oxides are formed when some substance combine with oxygen. These substances are either metals or non metal. And their combinations lead to four types of oxide

- Acidic oxide: some non metals like carbon, sulphur, phosphorus and nitrogen burn in oxygen to produce oxides called **acidic oxides**
- Basic Oxides: some metals like calcium, sodium, potassium and magnesium burns in oxygen to produce oxide called **basic oxides**
- Amphoteric oxides: These are exceptional metals that don't only behave as basic oxide like metals but also behave like acidic oxides. Examples are Zinc oxide, aluminum oxide, lead oxide and tin oxide
- Neutral oxides: they do not behave like acidic or basic oxides. Examples are carbon (iv) oxide, water, and nitrogen (ii) oxide.

Use of oxygen

- Oxygen is used in burning and respiration
- Oxygen is used to weld metals

Assignment

1. Give the chemical test to differentiate between hydrogen and oxygen

2. State three uses of hydrogen
3. Give four uses of oxygen
4. How can you test for oxygen in the laboratory?

MULTIPLE CHOICE QUESTIONS:

1. An electric bell is an example of an electromagnet
 - a. True
 - b. False
 - c. Maybe
 - d. I don't know
2. The lightest gas known to man is _____
 - a. Oxygen
 - b. Hydrogen
 - c. Fluorine
 - d. Halogen
3. The gas which is $1/15^{\text{th}}$ of air, useful to man and support burning is _____
 - a. Oxygen
 - b. Hydrogen
 - c. Fluorine
 - d. Argon
4. There are _____ types of water
 - a. 1
 - b. 2
 - c. 3
 - d. 4
5. The boiling point of water is _____ and its freezing point is _____
 - a. 100°C , 10°C
 - b. 10°C , 100°C
 - c. 100°C , 0°C
 - d. 100°C , 100°C
6. A phenomenon in which light rays bends as it crosses the boundary between two media of different densities is called _____
 - a. Interference
 - b. Polarization
 - c. Reflection
 - d. Refraction
7. Mechanical energy can further be subdivided into _____.
 - a. chemical and heat energy
 - b. gravitational and nuclear energy
 - c. kinetic and potential energy

- d. solar and wind energy
- 8. Energy is the _____.
 - a. Ability to do work
 - b. Capability to work hard
 - c. Force of gravity
 - d. Ability to resist work
- 9. The unit of energy is the _____.
 - a. Joules
 - b. Ampere
 - c. Newton
 - d. Metre
- 10. The following are examples of potential energy **except** _____.
 - a. Magnet at rest in a magnetic field
 - b. A coiled spring when stretched or compressed
 - c. A car in motion
 - d. A man waiting for a bus at the bus stop

WEEK 8: WATER

Objective: At the end of the lesson, student should be able to:

- i. List the two types of water
- ii. Explain the decomposition of water by electrolysis
- iii. Differentiate between temporary and permanent hard water
- iv. Explain different types of solutions

Oxygen react with hydrogen to produce water. There are two types of water (i) Natural water (ii) Treated water. Example of natural water is sea, lake, ocean, well, spring and rain water.

Natural water is water that exists naturally in its location. It is not manufactured or made in the laboratory or with any gas. Treated water is subjected to certain processes to make it useful for different purpose such as drinking and cooking. Reason for treating water is to make it germ free and safe to drink or cook

Properties of water

- Water is a colourless and tasteless when free of any chemical.
- Water is a universal solvent, this means water dissolve almost all solute
- It is transparent
- It is neutral to litmus paper
- The boiling point of pure water is 100°C and freezes at 0°C

Hardness of water

Water is said to be hard when it cannot produce lather easily with soap. This is caused by the presence of chemicals of calcium or magnesium dissolved in it these chemicals can be:

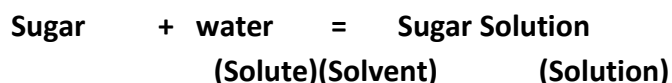
- Calcium (ii) hydrogen trioxocarbonate (iv)
- Magnesium hydrogen trioxocarbonate (iv)
- Calcium (ii) tetraoxosulphate (vi)
- Magnesium tetraoxosulphate (vi)

Types of hard water

- Temporary hardness: Temporary hardness of water contain either the first two, it can be removed by **boiling and addition of slaked lime**
- Permanent hardness: water is permanently hard when there is dissolved calcium (ii) tetraoxosulphate (vi) or magnesium tetraoxosulphate (vi). Such hardness cannot be removed by boiling but the following chemicals can be added to remove the hardness
 - (a) A substance known as permutite
 - (b) Washing soda i.e sodium trioxocarbonate (iv)
 - (c) Addition of caustic soda i.e sodium hydroxide.

Solutions

A solution is formed when a solute mixes uniformly with a solvent to form a solution. An example is when sugar (solute) dissolves in water (solvent) to form a sugar solution. This means that the solid that dissolves in a liquid is called a solute, e.g. sugar, and the medium through which the sugar is dissolved is called a solvent, e.g. water



Types of solution

There are about five types of solutions; they are summarized in the table below.

S/N	Type	Meaning	Example
1	True solution	Particles dissolve completely in the solvent such that you cannot see any particles with the naked eye	Sodium chloride solution
2	Colloidal solution (False solution)	Particles are larger than that of true solution but not large enough to be seen with the naked eyes. When a microscope is used, the particles are seen scattered in the container and they look cloudy in a beam of light.	Starch solution
3	Saturated solution	A solution that will dissolve any more solute, i.e. it is full and cannot take more solids. The opposite of this is called an unsaturated solution.	
4	Supersaturated solution	It contains more of the solute than it can dissolve at a particular temperature.	
5	Suspension	This is when there are undissolved particles in a solvent. The particles are so large as to be seen with the naked eyes. The particles settle at the bottom of the container or left standing and undisturbed.	Mixture of chalk, dust and water, muddy water

Assignment

1. Explain any 3 types of solution and give one example each.
2. Water is a universal solvent. Explain
3. What is hardness of water?
4. A liquid is labeled X. How can you prove that it is water?

MULTIPLE CHOICE QUESTIONS:

1. There are _____ types of water
e. 1

- f. 2
 - g. 3
 - h. 4
2. The boiling point of water is _____ and its freezing point is _____
- e. 100°C, 10°C
 - f. 10°C, 100°C
 - g. 100°C, 0°C
 - h. 100°C, 100°C
3. A phenomenon in which light rays bends as it crosses the boundary between two media of different densities is called _____
- e. Interference
 - f. Polarization
 - g. Reflection
 - h. Refraction
4. Mechanical energy can further be subdivided into _____.
- e. chemical and heat energy
 - f. gravitational and nuclear energy
 - g. kinetic and potential energy
 - h. solar and wind energy
5. Energy is the _____.
- e. Ability to do work
 - f. Capability to work hard
 - g. Force of gravity
 - h. Ability to resist work
6. The unit of energy is the _____
- e. Joules
 - f. Ampere
 - g. Newton
 - h. Metre
7. The following are examples of potential energy **except** _____.
- e. Magnet at rest in a magnetic field
 - f. A coiled spring when stretched or compressed
 - g. A car in motion
 - h. A man waiting for a bus at the bus stop
8. The formula for potential energy is _____
- a. Mgh
 - b. $\frac{1}{2}mv^2$
 - c. $\frac{1}{2}mgh$
 - d. Mv^2
9. A fruit from a tree of mass 20kg is 5m above the ground. What is the P.E. of the fruit above the ground. ($g = 10\text{m/s}^2$)
- a. 38.5j
 - b. 50.0j
 - c. 100j
 - d. 120j

10. A body falling under gravity will possess _____.

- a. Potential and kinetic energy
- b. Chemical and potential energy
- c. Pivotal and kinetic energy
- d. None of the above

WEEK 9&10: RADIOACTIVITY

Objective: At the end of the lesson, student should be able to:

- i. Define radioactivity.
- ii. List and describe the three types of radiation.
- iii. Identify the sources of radioactivity.

State at least two uses and two harmful effects of radioactivity. Radioactivity is the emission of radiation (transmission of energy in the form of wave or particle through space or through a material medium) from an unstable **atomic nucleus**. This energy is called **radioactive decay**. The radiation can be emitted in the form of a positive charged **alpha particle (α)**, a negative charged **beta particle (β)**, or **gamma rays (γ)**

Hazardous nature of radioactivity

- Radiation can alter the structure of cells in our bodies causing mutations which can produce cancer
- The more radiation a person is exposed to, the more dangerous it is. A dose of 100 resms will have probably cause some initial signs of radiation sickness such as loss of white blood cells, nausea, vomiting, and headache. With 300 rem dose, a person may lose hair temporarily, nerve cells and those that line the digestive tract will be damaged.

Uses of radioactivity

- Smoke detectors
- Security devices
- Food irradiation
- Industrial uses

Assignment

1. Mention three hazardous nature of radiation
2. Name the forms in which radiations are emitted
3. Aside the examples given in your note, name other uses of radioactivity